INTERNATIONAL ANALYTICS, AI, AND ALTERNATIVE ENERGY CONFERENCE 2026
DE LA SALLE UNIVERSITY

Instructional Supervision System

A Data-Driven Automated System for Monitoring and Evaluation of Teachers' Performance

Engr. Reynel Mario Canor
La Salle Green Hills, Philippines

Act as a subject coordinator. Review every teacher's curriculum map, learning guide and classroom observation for three trimesters. Then rate each one fairly for merit, tell each exactly what to improve, and find your department's training needs. Show your evidence.

Put on your Subject Coordinator hat.

01 · OPENING

How do we keep up?

Many teachers. Many evaluation documents. Different sets of performance indicators.

?

How do we keep track of everything?

?

How do we see patterns across teachers and departments?

?

How do we turn records into feedback and development?

01 · WHY IT MATTERS

Evaluation shapes a teacher's growth — and pay

RA 10912

The Philippine CPD Act requires teachers to keep upgrading — which needs **objective** evidence of the gaps.

11 of 19

countries in an OECD appraisal study link evaluation results to **salary**.

Sources: Republic Act 10912 (2016); OECD (n.d.); Darling-Hammond et al. (2017)

02 · THE PROBLEM

Three aspects are evaluated

Curriculum Map

How learning outcomes are planned and organized

Learning Guide

How the curriculum becomes instructional design

Classroom Observation

How instruction is implemented in the classroom

But the data was handled **manually and in isolation.**

02 · THE PROBLEM

The problem was not the absence of data. It was how we managed, connected, and used the data.

- Performance trends were hard to see
- Training needs were hard to pinpoint
- Merit rating was slow to consolidate

Anchored on alignment, built by iteration

THEORY

Constructive Alignment (Biggs)

Intended outcomes, learning activities and assessment must align.

Planned → Designed → Implemented

METHOD

Design-Based Research

Refined through actual use, user feedback and emerging requirements.

Not a one-time product

Three connected subsystems

PLANNING

CM Analytics

4 areas: inquiry, concept-building, transfer, self-evaluation

28

INDICATORS

DESIGN

LG Analytics

9 instructional designs, ID1 to ID9

34

INDICATORS

IMPLEMENTATION

COT Analytics

13 teacher and 7 student indicators

20

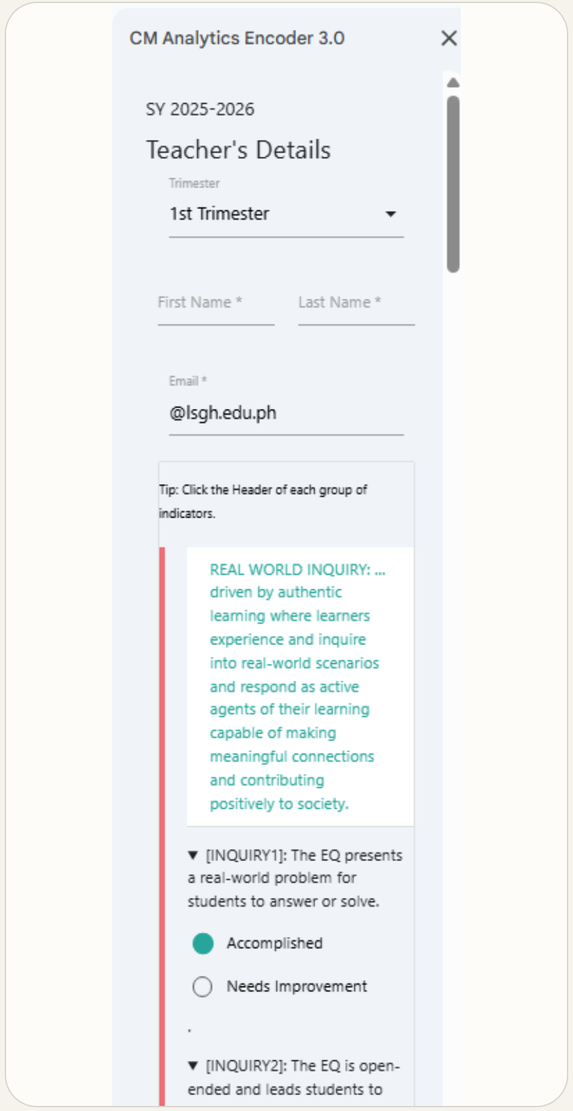
INDICATORS

Connected, not separate: one view of performance from planning to classroom implementation

04 · GOOGLE SHEETS + APPS SCRIPT

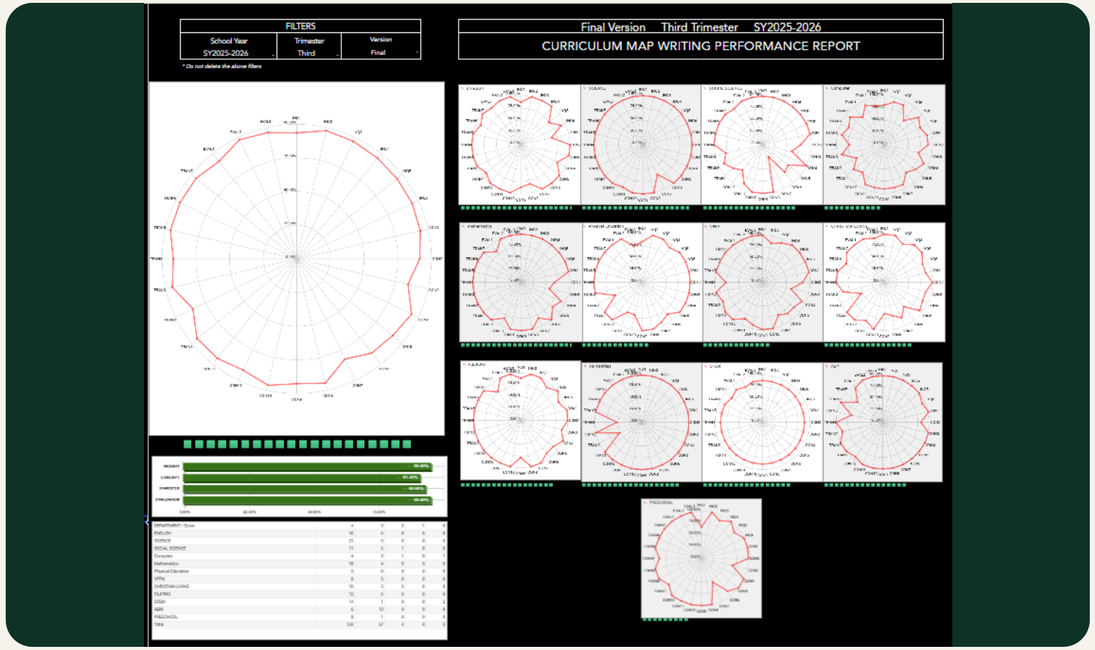
One click turns a checklist into a report

ENCODE	Mark each indicator **Accomplished** or **Needs Improvement**
COMPUTE	Scores on a **four-point** system, stored automatically
VISUALIZE	The analytics **dashboard** updates
REPORT	A **PDF report** goes to the teacher via Gmail



05 · ANALYTICS AND REPORTING

The dips are the training needs



CM and LG radars: every indicator at once, by department



COT dashboard: gauges, look-fors and a tracker — 99.55% observed

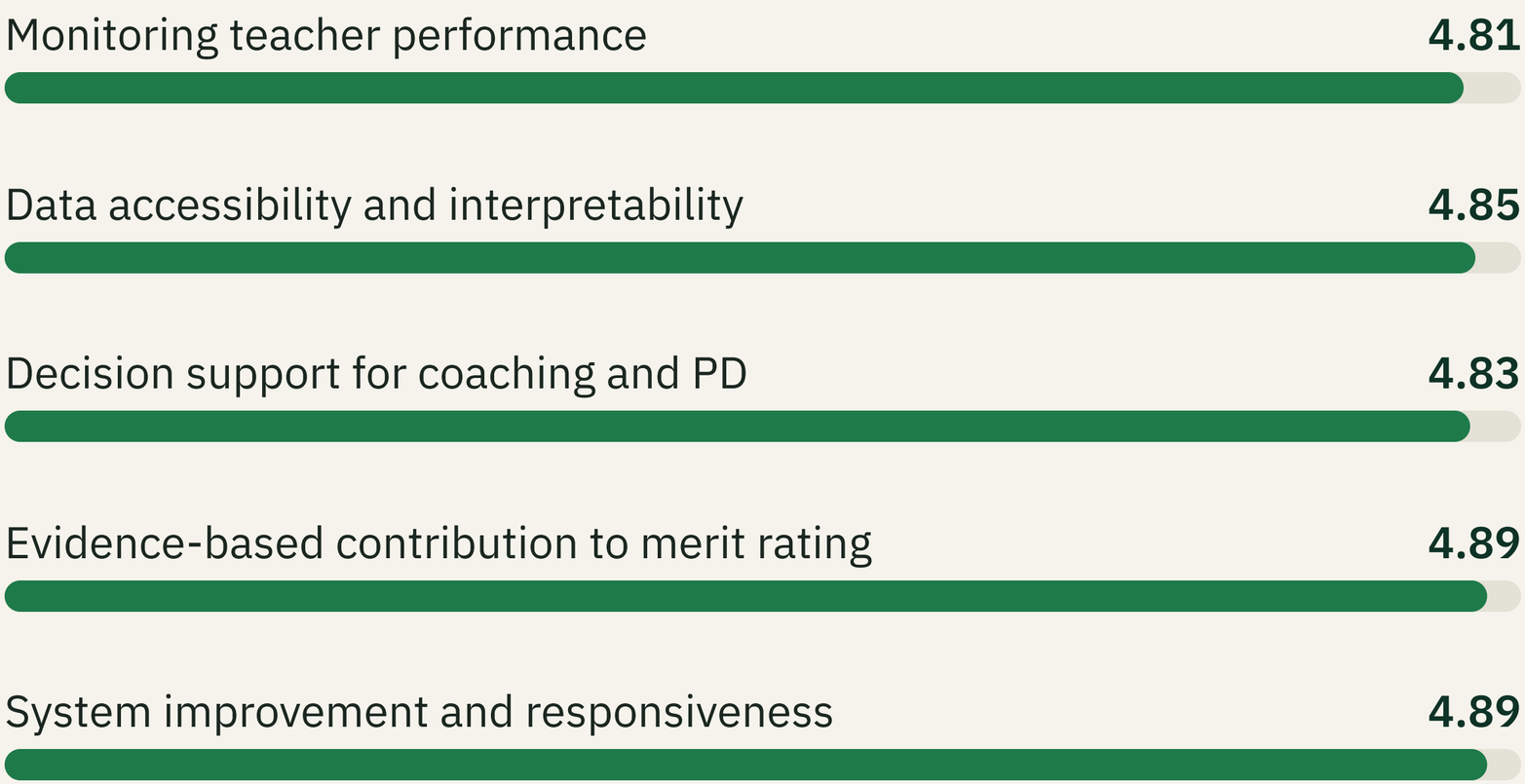
05 · MERIT RATING

Evidence becomes one merit score

CM Analytics	**Domain 1** · Curriculum Planning and Development	5%
LG Analytics	**Domain 2** · Content Knowledge and Pedagogy	5%
COT Analytics	**Domain 3** · Instruction	15%

Averaged across **three trimesters** — one basis for merit rating.

Every construct rated above 4.8



10

Subject Coordinators surveyed

Five-point Likert scale

Beyond the numbers

In coordinators' words

"...ang kalakasan at kailangan pang paunlarin ng mga guro at ng department."

the strengths of teachers and the department, and what still needs developing

"...design more targeted and specific action plans."

Six development cycles refined

- Scoring and visualization
- Indicator-level analysis
- Data integrity
- Collaborative evaluation
- Automated report distribution

Key contribution: a school can grow its own decision-support system by answering real operational needs.

07 · CONCLUSION

Three contributions

Act as a subject coordinator. … Show your evidence. Now you can.

1

A **systematic, measurable** way to monitor teacher performance

2

Professional development needs identified through specific indicators

3

A structured, **evidence-based** contribution to merit rating

07 · LIMITATIONS AND RECOMMENDATIONS

Limits and next steps

Limitations

- Criteria are institution-specific
- Coordinators use it; teachers receive reports
- Teacher-made assessments not covered
- Effect of training interventions not measured

Recommendations

- Keep calibrating the scoring model
- Stronger access controls
- Standardized teacher identification
- Report preview before sending
- Periodic review of indicators

08 · CLOSING REMARKS

The real value of evaluation is what we do with the information afterward.

From a compliance activity to an instrument for continuous improvement.

Thank you very much.

Engr. Reynel Mario Canor · La Salle Green Hills

NEXT STEP · LEARNING OUTCOMES ANALYTICS

Evaluation completes the cycle

1 · PLANNING

CM + LG Analytics

Is the plan aligned with intended outcomes?

In use

2 · IMPLEMENTATION

COT Analytics

Is the plan carried out in the classroom?

In use

3 · EVALUATION

Learning Outcomes Analytics

Did students actually learn? Results feed back into planning.

Next

Open Learning Outcomes Analytics

LSGH Google account required